32933 research project- Type 1 prototype
The UTS Project Match System
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Executive Summary

The core purpose of AI Project Match is to streamline the student project selection process by integrating AI to personalize recommendations. Students can input their interests in natural language, and the system uses DeepSeek API to analyze their input and match them with relevant projects from database. Both students and teachers can register and log in, with role-based access determining available actions. Teachers can create and edit projects, while students can create and view their own projects. The platform adopts role-based functionality, where students and teachers access different functions through a shared interface structure.

The system was implemented using Flask as the backend framework, SQLite as the database, and Tailwind CSS for responsive design. SQLAlchemy was used to perform database operations. The AI integration involved a two-stage prompt strategy: the first to extract project-related requirements (fields, skills, keywords), and the second to rank available projects based on relevance. This approach simplified natural language input into structured data used for intelligent project suggestions.

During testing, we validated both backend functionalities and user interface behavior. The registration, login, project creation, project editing, and AI recommendation flows were tested successfully using predefined accounts.

Several challenges were encounterd, such as incorrect role routing and API errors due to invalid or missing credentials. These issues were resolved through code refactoring, UI adjustments, and better error handling.

While the AI provided reasonable matches, occasional failures due to API key issues or overly

generic user input highlighted the system's reliance on external services and the importance of input clarity.

Overall, the project demonstrates a well-executed integration of AI-driven functionality into a practical educational tool. It combines modular backend architecture with a lightweight frontend to deliver a clear and focused experience for both roles. The dual-stage use of the DeepSeek API provided accurate, structured recommendations, while the system's role-based logic and responsive interface ensured a smooth and intuitive user journey.

Table of Contents

Executive Summary 2

Introduction 5

Literature Review 5

Future optimization and improvement： 7

Proposed Solution 7

System Design Philosophy 8

Technical Implementation and Innovations 8

Conclusion and Future work 11

References 13

Introduction

In many IT education programs in colleges and universities, students often need to complete graduation projects, the process of selecting a suitable project can often be overwhelming, particularly for those with limited exposure to real-world problem domains.

To address this challenge, The AI Project Match platform was developed to simplify this process by providing students with personalized project recommendations based on their interests and technical backgrounds. At the same time, the platform also provides teachers with an interface for publishing and managing projects.

This report outlines the motivation behind the system, its design and architecture, its main features (including project creation and recommendations), test results, and conclusions drawn from the development process.

Literature Review

As Jeon explained, AI chatbots is considered as valuable in teaching and enhanced student's willing to chat with it (Jeon, 2024). Besides, Li and Coates found that AI chatbot like ChatGPT can efficiently provides resources that teachers and students need in their university library, their innovative system 'Libchat' can search in the library and return the books as required for referencing (Li & Coates, 2025). These advancing systems inspires us, can we build a similar system for students' research project match?

This project aims to provide a platform for students and teachers to choose graduation project

matching based on professional skills in the final semester. By comparing the existing system with the official UTSProjectMatch system, we can find that the original UTS system has student login and teacher login, and the following details the functions of each role, so that people can understand the responsibilities of each role at a glance. The page we designed is very simple, which cannot make people clearly understand their responsibilities, and we also have a registration function. Although in most cases this function can facilitate members who do not have an account to register, as a school's ProjectMatch, it should only be allowed to be used by internal school personnel. Giving registration permissions to outsiders may cause troublemakers, such as adding or deleting projects at will, confusing the public. (UTS, 2025)
On the student login page, the system we designed can use artificial intelligence to help students better choose projects of interest. When students consult artificial intelligence about relevant skills or topics of interest, artificial intelligence will give a response. For example, if you say, "I want to do a project about medicine", the system will automatically recommend matching projects. In addition, students can create projects by clicking "Create My Own Project", which gives students the authority to create their own projects and can send links to teammates and invite others to join. However, there is no function to match mentors to the created projects, which need to be improved in the future. In the student-side function of the original UTS official platform, students can register their graduation project topics and find available mentors. They can check whether the project is approved in the system, or if they don't know what they want to do, they can directly browse existing projects and apply to join.

On the teacher side of the UTS official platform, teachers can review applications submitted by students, determine whether the projects applied for by students meet the standards, and publish new project titles for students to choose from. When the student is determined to be in charge of the tutor, the tutor can directly view the status of the project he is in charge of and the project schedule in the system. In the teacher side we designed, teachers can also publish and edit their own projects in the system, and what is better than the official one is that each project will show its field and skill requirements, and you can also check which students are

interested in the project, but the system is not perfect yet, and there is still a lot of room for improvement.

Future optimization and improvement：

Optimize the logic of AI, provide AI with more data for training, and improve the user experience. Add permissions for email login and registration, enhance network security management, identify outsiders, and reduce external interference. Add the number of students that can be guided by a tutor to ensure that a tutor will not be responsible for too many projects, resulting in the teacher being unable to take care of everyone. When there are too many applicants, the applicants will be arranged according to the application time. Add project timetables and progress displays to better help students and tutors better understand the stage of the project and the project plan. Add email reminders for project approval or rejection on the student page. When a student applies for a project and decides on a tutor, an email will be sent to the tutor to remind him to prevent the tutor from missing the student's application. Add a button to cancel interest on the student page, and when AI recommends a project, it will give the reason for recommending the project and a brief description of the project.

Proposed Solution

The AI Project Match platform was conceived to address a recurring challenge in higher education and project-based learning: how to efficiently and intelligently match students with suitable AI-related projects. By leveraging modern AI models for both natural language understanding and recommendation, the system aims to bridge the knowledge gap between students' self-reported skills/interests and the diverse pool of available projects. This technical reflection discusses the rationale behind the architectural choices, the effectiveness of the dual-stage AI integration, the merits and drawbacks of the technology stack, and the prospects for further development.

System Design Philosophy

At its core, the platform is built on the principle of intelligent mediation—facilitating more meaningful connections between students and projects through automation and data-driven analysis. Unlike traditional project selection platforms, which often rely on keyword search or manual curation, AI Project Match incorporates semantic understanding and personalized ranking.

A key architectural decision was the adoption of a two-stage AI workflow. The first stage translates ambiguous student input into structured requirements, while the second stage evaluates and ranks projects based on these requirements. This separation allows for modularity and easier troubleshooting, as well as the potential for independently upgrading each stage as more advanced AI models become available.

Technical Implementation and Innovations

1. Dual-Stage AI Integration

The highlight of the system is the dual interaction with the DeepSeek API:

Requirement Extraction: In the first stage, the platform uses AI to parse free-form student input, extracting structured data such as fields, skills, and features. This not only mitigates the ambiguity inherent in natural language but also reduces the risk of bias or error from manual parsing.

Project Ranking: The second stage leverages AI to compare extracted requirements against all available projects, producing a ranked list with justifications. This approach is more sophisticated than basic filtering, as it allows the AI to weigh multiple criteria and contextual

factors.

An important technical insight was the use of controlled temperature settings in the AI API calls. Lower temperature ensures consistent, deterministic extraction of requirements, while a slightly higher temperature in the ranking phase promotes richer, more nuanced matching rationales.

2. Clean API Layer and Modular Design

The backend is implemented using Flask, providing a clean RESTful API (/api/chat) that abstracts away the complexities of AI interactions and database operations. Business logic, such as parsing AI responses and filtering projects, is neatly encapsulated in helper functions. This structure makes it easy to extend the platform with new endpoints or swap out AI providers.

The use of SQL Alchemy ORM for data persistence not only simplifies CRUD operations but also enforces data integrity through explicit model relationships. The class diagram reveals a normalized schema that supports efficient queries and future scalability.

The frontend eschews heavy frameworks in favor of vanilla JavaScript and utility-first Tailwind CSS. This approach reduces bundle size, eases onboarding for new contributors, and allows direct manipulation of the DOM for dynamic chat interactions. As a result, the system achieves a balance between interactivity and maintainability.

The platform is designed for rapid deployment and ease of use. Clear setup instructions, automated initialization of test accounts, and reliance on widely available tools (Python, SQLite) lower the barrier for new developers and testers. This is especially valuable in educational contexts, where the audience may have varying levels of technical expertise.

Semantic Matching: By leveraging AI for both extraction and ranking, the platform achieves a

high degree of alignment between student intent and project recommendations, outperforming simple keyword-based systems.

Extensibility: Modular architecture and clean code separation enable easy addition of features, such as support for more user roles or alternative AI providers.

User Experience: The chat-based interface, coupled with real-time updates and attractive project cards, makes the process engaging and intuitive.

Dependence on External AI APIs: The platform's intelligence hinges on the reliability and accuracy of third-party AI services. Any fluctuations in API quality or availability can directly impact user experience.

Cold Start Problem: The effectiveness of the system is limited by the breadth and quality of project data. With few projects or limited diversity, recommendations may lack relevance.

Potential for Overfitting: Hardcoded thresholds (e.g., score ≥ 3) and prompt templates may not generalize well across different cohorts or subject areas.

Security and Privacy: Storing user data and integrating with external APIs raises concerns about data protection, especially in educational settings.

Through iterative development and testing, several valuable lessons emerged:

Prompt Engineering is Critical: Subtle changes in AI prompt wording can significantly affect output quality. Consistent formatting and explicit instructions to the AI model are essential.

Error Handling Must Be Proactive: Given the unpredictability of AI responses, fail-safe and clear error messages are necessary to prevent system crashes and user frustration.

User Feedback Loops are Invaluable: Collecting and analyzing user feedback (e.g., which projects students actually select) can inform future refinements to both the AI models and ranking criteria.

Looking ahead, several enhancements are envisioned:

Adaptive Learning: Incorporating feedback to refine AI prompts and ranking logic, making the

system smarter over time.

Support for Group Projects: Extending matching logic to teams rather than individuals, accounting for collective skill sets and preferences.

Multi-language Support: Broadening accessibility by enabling requirement extraction and project descriptions in multiple languages.

Enhanced Security Measures: Implementing OAuth, role-based access control, and encrypted storage for sensitive data.

The AI Project Match platform represents a thoughtful synthesis of modern AI capabilities and pragmatic web development. Its dual-stage AI workflow, modular backend, and lightweight frontend together provide a compelling solution to a longstanding problem in project-based education. While challenges remain—particularly around data quality and external dependencies—the platform's open, extensible design offers a strong foundation for ongoing innovation and real-world impact.

Conclusion and Future work

In conclusion, this report illustrates the process of building a student project match system. The platform allows students to create projects and ask AI to rank the project close to its requirements, students can also select to join a project and leave the project. For supervisors, they can create new projects, and they have a dashboard listing all the projects created by them. Besides, both two kinds of users can register and login. In the registering session, the user can choose his role, with access control based on user role and encrypted user token, system can make sure that a user can only access the function related to their role to achieve the authentication and authorization.

Above all, our group provides a simple, useful and scalable foundation for a more complicated project match system. In future developments, there are other functions can be added or improved, such as:

- More research fields
- User detail editing
- User password resetting
- Student applies for a project
- Supervisor quota
- Supervisor approval for a project created by students
- The coding style should comply with the established development guidelines. Our functions are concentrated in one single file, which is not low coupling or high cohesion, the project index is also a big problem.

Besides, the pipeline of the project can and should be improved, this is because our group has a chaotic project schedule management, backend finished quickly but front-end teams confused on understanding some parameters, the name of parameters are not unified, backend finished while the front end did not start. We have a UML diagram guideline in the beginning, but some group members do not have a concept of it, which results in editing the structure of the project. We should have pair-programming each week, but we failed. If we can finish some parts simultaneously, the front end and back-end work together and implement a few functions week by week, that would be good practice. Overall, we should have better project management and time management in future.

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